

Technology News

Robot that walks on water

Scientists in China have created a microbot that imitates the water-walking capabilities of the 'water strider'—a long-legged insect that effortlessly skims across pond and lakes surfaces. The aquatic robot weighs as much as 390 water striders but does not sink when placed on water. In fact, it stands, walks and runs effortlessly on its surface.

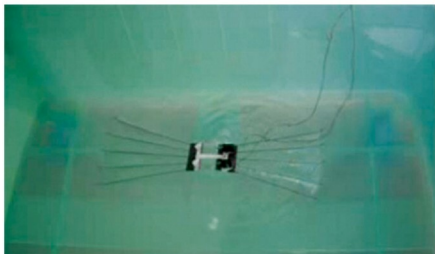
Scientists claim that the bionic bot is a vast improvement over previous devices of this kind, positioning it as a prime candidate for military spy missions, water pollution monitoring and other applications. It has ten water-repellent, wire legs and also two movable, oar-like legs propelled by miniature motors.

According to Qinmin Pan, associate professor of chemical engineering, Zhejiang University, who has achieved this rare feat, these robots are not only practical and agile but also inexpensive.

Nanovision possible through X-ray microscope

We can make things at nanoscale, but can't see them very well. Researchers at University of California have now developed a microscope that can penetrate deep within materials and see details that are as small as a billionth of a metre. This microscope does not even use a lens. A powerful computer program converts patterns from X-rays bouncing off materials into images of objects as small as one nanometre across, on the scale of a few atoms.

According to study researcher Oleg Shpyrko, the technology could be utilised to look at different elements inside a material, or to image viruses, cells and tissue in great detail. But the most important application will be nano-sized engineering.



The water strider robot in action (Image courtesy: American Chemical Society)

Bionic fingers

Chloe Holmes, 15, is the youngest person in Europe to be fitted with bionic fingers. She had lost all her fingers to septicaemia in her childhood. A full

set of bionic left digits costing Rs 2.86 million (£38,000) have been fitted to her hand which are controlled by electronic signals from the nerve endings in her hand.

The bionic fingers, which have been made specifically to fit Holmes' hand, were made by Touch Bionics in Scotland. The hand is fitted with sensors in the sleeve which enable her to control the digits.

Holmes is now learning basic tasks such as cleaning her teeth and using a knife and fork for the first time. She said, "The best thing about my new hand is that now I can hold things."



ProDigits

Gold particles to boost organic solar cells' efficiency

While organic photovoltaic solar cells are much thinner and less expensive to produce than conventional solar cells made with inorganic silicon wafers, they still lag behind in their ability to efficiently convert sunlight into electricity.

Now UCLA researchers and their colleagues from China and Japan have shown that incorporating